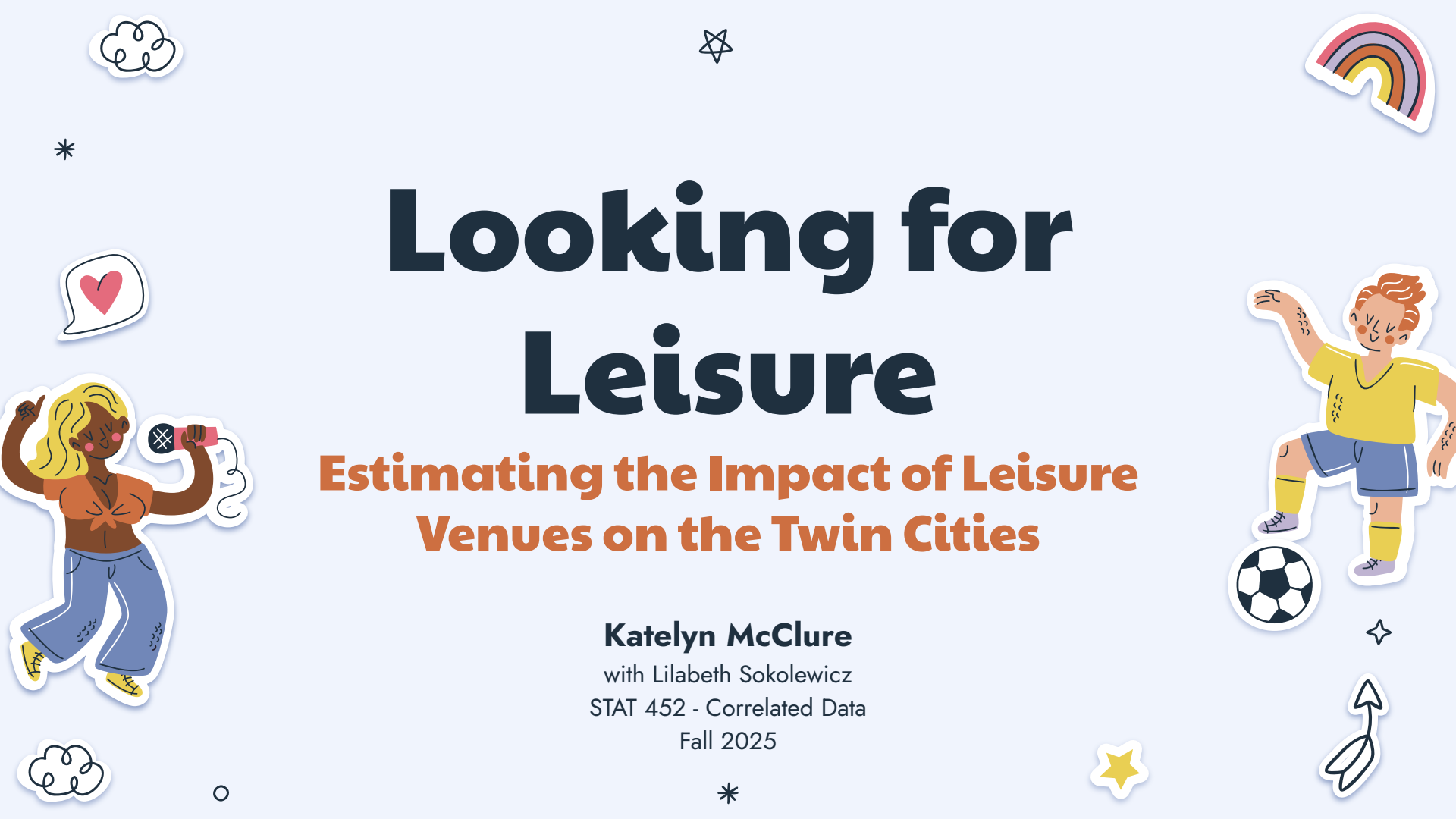


Looking for Leisure

**Estimating the Impact of Leisure
Venues on the Twin Cities**

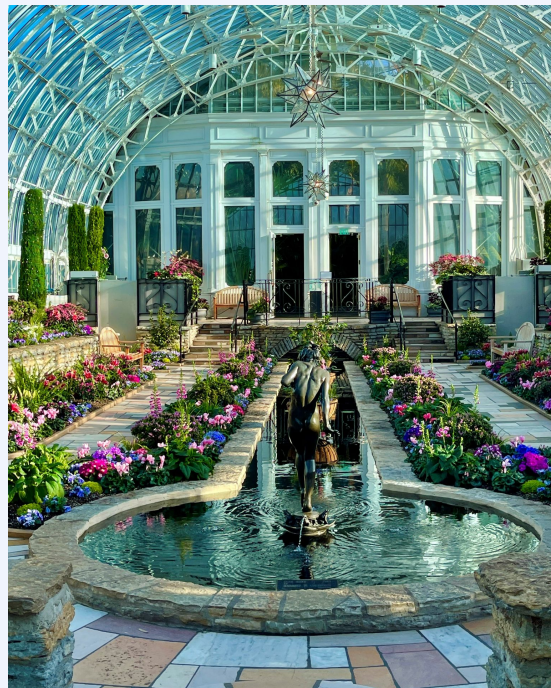
Katelyn McClure

with Lilabeth Sokolewicz
STAT 452 - Correlated Data
Fall 2025





Target Field
Minneapolis



**Como Zoo
Conservatory**
Saint Paul

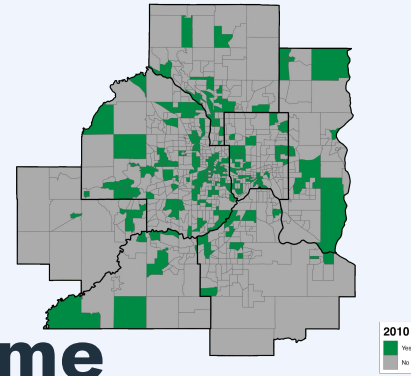


Grandview Theater
Saint Paul





Research Question



How are **property value** and **income** related to the presence of ≥ 1 leisure venue in a Twin Cities census tract, and how does that **change over time** between 2010 and 2020?

Yes/No Leisure Venue $\sim$ Year + Median Household Income + Average Total Property Value





Yes/No Leisure Venue ~ Year + Median Household Income + Average Total Property Value

Data

Leisure Venues

- museums
- performance theaters
- movie theaters
- zoos
- aquariums
- public gardens
- professional sports venues*

*NaNDA, Wikipedia**

Census Tract

~4,000 residents

Property Value

Average Land + Building Value

MetroGIS

Income

Median household income

US Census





Spoonbridge and Cherry
Minneapolis
2024

Background

Leisure Venues - pros and cons

Community-based Arts

- Health and well-being
- Social cohesion
- Increased community involvement
- Improved quality of life
- Economic benefits



Chanhassen Dinner Theatres
Chanhassen
2019



Location Matters



Minnesota Zoo
Apple Valley
2005

- Nearness to venue highly predictive of attendance
- Fewer transportation barriers
- Nearby = more likely to reap benefits

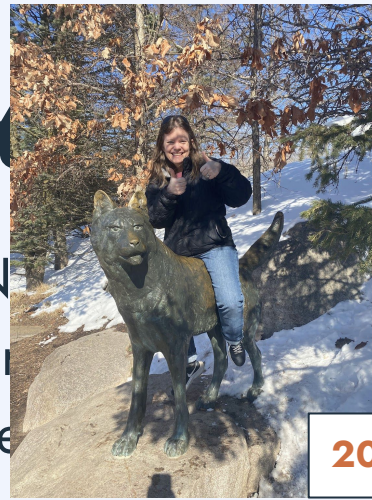




2011



2015



2023



2019



2025



2023

Gentrification

- Commercial arts venues
- Fast-changing areas
- Attract higher-income residents
 - Often white





Displacement



- Increase nearby property values
- Increase rent
- Displace marginalized communities



Interstate 94 Construction

1940 2020





Fitzgerald Theater
Saint Paul
2025

Model

Longitudinal Data, GEE



How are **property value** and **income** related to the presence of ≥ 1 leisure venue in a Twin Cities census tract, and how does that **change over time** between 2010 and 2020?



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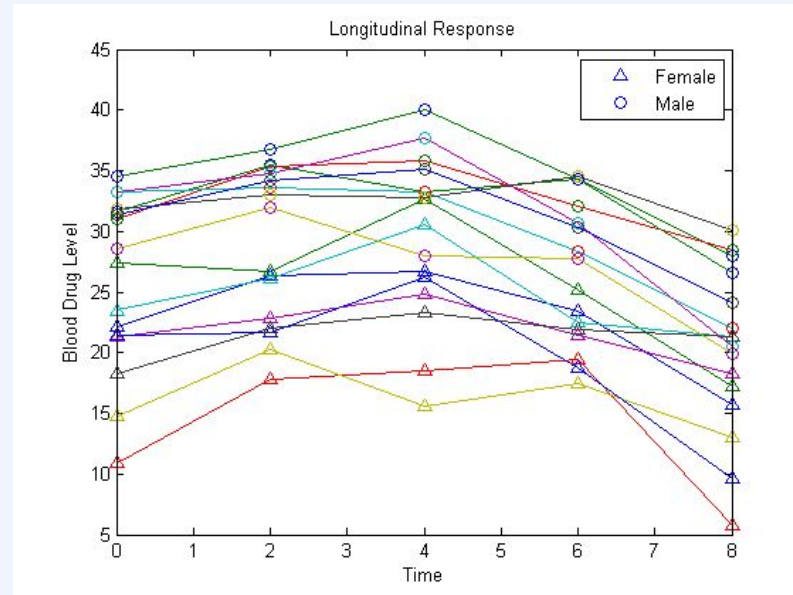
○

Yes/No Leisure Venue ~ Year + Median Household Income + Average Total Property Value



Longitudinal Data

- ★
 - Repeated measures on same tracts over time
 - Correlated, not independent
- $\text{lm}(y \sim x + \dots, \text{data} = \dots)$
 - Assumes independence
 - Standard Errors



Longitudinal Data

Same subjects over time

Visual unrelated to Leisure Venues

○





Yes/No Leisure Venue ~ Year + Median Household Income + Average Total Property Value



GEE

```
> gee_mod %>% summary()
```

	Estimates	Model SE	Robust SE	wald	p
(Intercept)	-1.4980	0.21180	0.21630	-6.925	0.000e+00
I(avg_total_val/1e+05)	0.3088	0.04205	0.04808	6.423	0.000e+00
I(IncomeE/1e+05)	-1.0210	0.23590	0.25060	-4.073	4.638e-05
as.factor(year)2020	0.3444	0.11030	0.11200	3.076	2.098e-03

○



Allianz Field
Saint Paul
2019

- Generalized Estimating Equations
- Logistic regression
- Clustering unit
- Correlation structure
 - Exchangeable
- Robust Standard Errors
- Population Estimate

○



Model

Year

2010, 2020

+

**Median
Household
Income**

Per \$100,000

+

**Average
Property
Value**

Average land value +
average building value
Per \$100,000

~

Contains Leisure Venue?

Yes/No



Science Museum of Minnesota
Saint Paul
2011

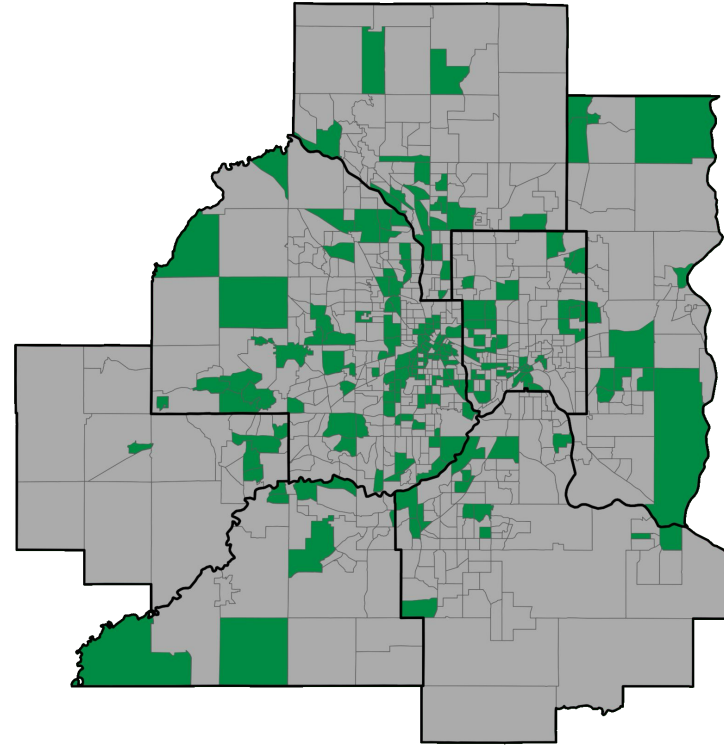
Results

Not what we'd expect...

Year

41% increase
in odds

Leisure Venue Presence
2010 vs 2020



2010
Yes
No

Property Value

36% increase
in odds



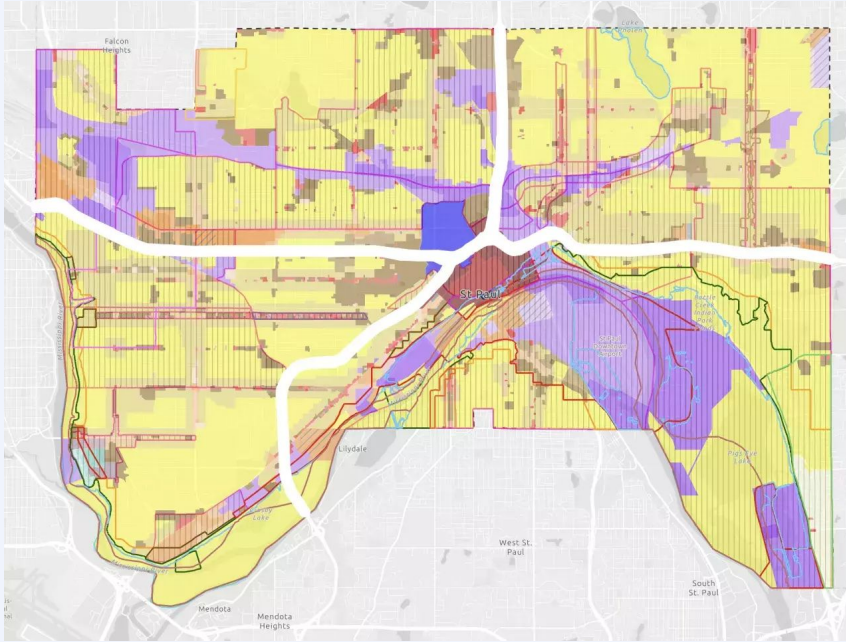
Orpheum Theatre
Minneapolis
2009



Target Field
Minneapolis
2010

Income

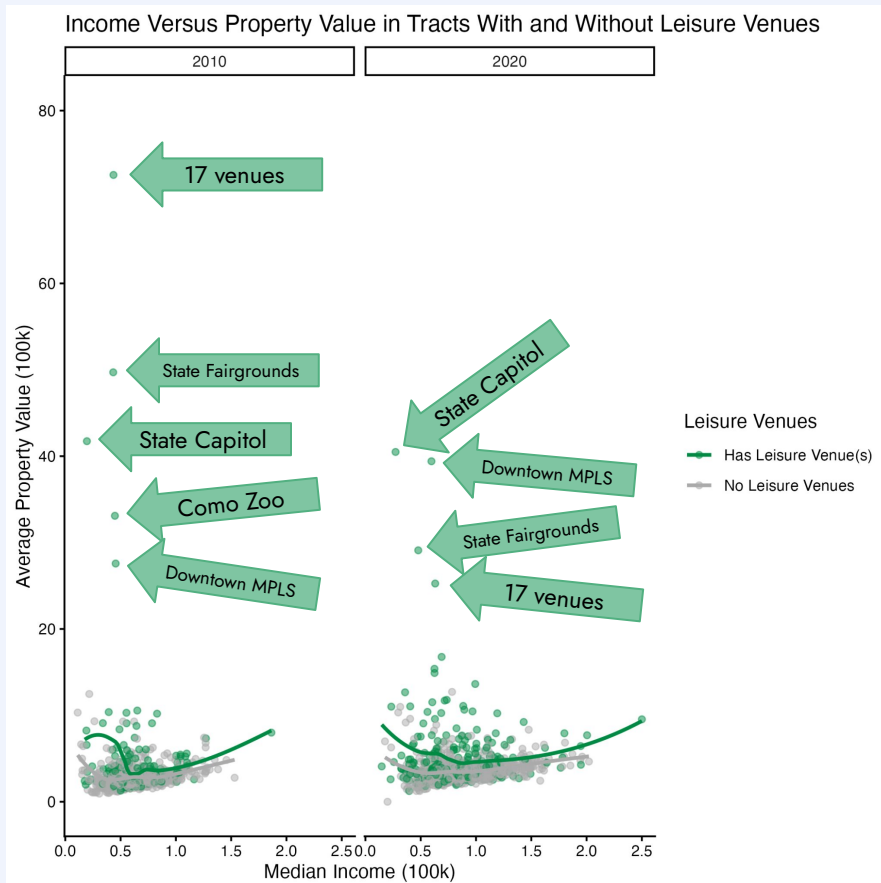
**64% decrease
in odds**



Saint Paul Zoning
2025



Income vs Property Value





Minnesota Landscape Arboretum
Chaska
2018

Conclusions

Conclusions

- More people are closer to venues in 2020
 - Good? Bad?
- Venues increase nearby land values
 - Good? Bad?
- Low income tracts more likely to contain venues
 - Good? Bad?



Xcel Energy Center
Saint Paul
2025
Go Frost



Limitations



CHS Field
Saint Paul
2024

- Two time points
 - 2010, 2020
- Don't distinguish between arts & sports
- Commercial status
- Metro-averaged estimates
- Narrow definition of leisure venue





Thank You!



Macalester College
Saint Paul
2022

